• Further, note two "delicate" aspects in this sentence. First, the number N should exist for any positive number ε. Obviously, there is an infinite set of such ε. Second, inequality (1) should hold always (i.e. for each ε) for all $n > N$. But there is an equally infinite set of numbers n!

Reader: Now, the definition of the limit has become more obscure.

Author: Well, it is natural. So far we have been examining the definition "piece by piece". It is very important that the "delicate" features, the "cream so to say, are spotted from the very outset. Once you understand them, everything will fall into place.

• In fig 7a there is a graphic image of a sequence. Strictly speaking, the first 40 terms have been plotted on the graph. Let us assume that if any regularity is noted in these 40 terms, we shall conclude that the regularity does exist for $n > 40$. Can we say that this sequence converges to the number a (in other words, the number a is the limit of the sequence)?

Reader: It seems plausible.

Author: Let us, however, act not on the basis of our impressions but on the basis of the definition of the limit of sequence. So, we want to verify whether the number a is the limit of the given sequence. What does the definition of the limit prescribe us to do?

Reader: We should take a positive number ε.

Author: Which number?

Reader: Probably, it must be small enough.

Author: The words "small enough" are neither here nor there. The number ε must be $\boxed{\text{arbitrary}}$.

Thus, we take an arbitrary positive ε. Let us have a look at Fig. 7 and lay off on the y-axis an interval of length ε, both upward and downward from the same point a. Now, let us draw through the points $y = a + \varepsilon$ and $y = a - \varepsilon$ the horizontal straight lines that mark an "allowed" band for our sequence. If for any term of the sequence inequality (1) holds, the points on the graph corresponding to these terms fall inside the "allowed" band.